A first-order primal-dual algorithm for convex problems with applications to imaging

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Introduction

Paper: Antonin Chambolle, Thomas Pock. *A first-order primal-dual algorithm for convex problems with applications to imaging*. 2010. ⟨hal-00490826⟩

Problem:

- Variational methods are essential for ill-posed inverse imaging problems.
- While total variation regularization preserves sharp discontinuities (edges), it is difficult to minimize due to the non-smoothness of the functional.
- The authors introduce a flexible first-order primal-dual algorithm specifically designed for structured, non-smooth convex optimization problems.
- Unlike prior schemes that either required expensive linear system solves per iteration or stalled at slow subgradient rates, this method achieves an optimal $O(1/N)$ convergence rate for general non-smooth problems.
- The algorithm accelerates to $O(1/N^2)$ when either the primal or dual objective is uniformly convex, and achieves linear convergence $O(1/e^N)$ when both are uniformly convex.

Optimization Problem

$$\min_{x \in X} \; F(Kx) + G(x)$$

such that:

- X, Y: finite-dimensional real vector spaces.
- $K : X \to Y$ is continuous, linear; $\|K\| = \max\{\|Kx\| : x \in X, \|x\| \leq 1\} < \infty$.
- $G : X \to [0, \infty)$ and $F : Y \to [0, \infty)$ are proper, convex, l.s.c. functions.

By **Fenchel-Legendre duality**, $F(Kx) = \max_{y \in Y} \langle Kx, y \rangle - F^*(y)$, converting the primal problem into the **saddle-point problem**:

$$\min_{x \in X} \max_{y \in Y} \langle Kx, y \rangle + G(x) - F^*(y)$$

Assuming:

- There exists saddle point $(\hat{x}, \hat{y})$ satisfying:

$$K\hat{x} \in \partial F^*(\hat{y}), \quad -K^*\hat{y} \in \partial G(\hat{x})$$

- F, G are "simple": resolvents $(I + \tau \partial F)^{-1}(y) = \arg\min_x \left\{ \frac{\|x-y\|^2}{2\tau} + F(x) \right\}$ and $(I + \tau \partial G)^{-1}$ have closed-form solutions or can be solved efficiently.

Main Algorithm

Initialization: Choose $\tau, \sigma > 0$ with $\tau\sigma\|K\|^2 < 1$, $\theta \in [0,1]$, $(x^0, y^0) \in X \times Y$, and set $\bar{x}^0 = x^0$.

For each iteration $n \geq 0$:

1. **Dual Proximal Ascent Step (y^{n+1}):**
   - Forward gradient ascent along the coupling term $+\sigma K\bar{x}^n$ to maximize with respect to y via proximal minimization.

$$y^{n+1} = (I + \sigma\partial F^*)^{-1}(y^n + \sigma K\bar{x}^n) = \arg\min_{y \in Y}\left\{\frac{\|y - (y^n + \sigma K\bar{x}^n)\|^2}{2\sigma} + F^*(y)\right\}$$

2. **Primal Proximal Descent Step (x^{n+1}):**
   - Backward gradient descent along the adjoint direction $-\tau K^* y^{n+1}$ to minimize with respect to x via proximal minimization.

$$x^{n+1} = (I + \tau\partial G)^{-1}\left(x^n - \tau K^* y^{n+1}\right) = \arg\min_{x \in X}\left\{\frac{\|x - (x^n - \tau K^* y^{n+1})\|^2}{2\tau} + G(x)\right\}$$

3. **Linear Extrapolation ($\bar{x}^{n+1}$):**

$$\bar{x}^{n+1} = x^{n+1} + \theta(x^{n+1} - x^n)$$

Acts as an approximate extragradient step along the primal velocity direction to anticipate the next position, algebraically canceling the cross-coupling error without an implicit subproblem.